Report Two

School Management System

Project Title: School Management System

Report Title: UML Class Diagram Modelling

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Introduction

The School Management System is a software system designed to organize and manage important school activities. It helps manage student information, teacher details, courses, and student enrollments. This report presents the UML class diagram of the proposed system, including its main classes, attributes, methods,

and relationships.

Description of the Main Classes

The School Management System consists of four main classes: Student, Teacher, Course, and Enrollment. Each class represents an important part of the system and has specific responsibilities.

Student Class

The Student class represents students registered in the school system. It stores student information and allows student details to be updated.

The Teacher class

represents teachers in the school system. It stores teacher information, including their names and specializations, and supports course assignment.

Course Class

The Course class represents the courses

offered by the school. It stores course information, such as the course name and number of credits, and allows course details to be managed.

Enrollment Class

The Enrollment class manages the registration of students in courses. It stores enrollment information, including the enrollment date and status.

Attributes and Methods

Each class in the School Management System contains attributes and methods that define its data and behavior. Attributes represent the information stored by each class, while methods represent the operations that the class can perform.

Student Class

Attributes: studentId, fullName, email.

Methods: register(), updateDetails().

Teacher Class

Attributes: teacherId, fullName, specialization.

Methods: assignCourse(), updateDetails().

Course Class

Attributes: courseId, courseName, credits.

Methods: addCourse(), updateCourse().

Enrollment Class

Attributes: enrollmentId, enrollmentDate, status.

Methods: enrollStudent(), updateStatus().

Relationships and Multiplicity

The UML class diagram shows the relationships between the main classes in the School Management System. Multiplicity indicates how many instances of one class can be associated with instances of another class.

Student and Enrollment

One student can have zero or many enrollments, while each enrollment belongs to one student.

Multiplicity: Student (1) — Enrollment (0..*)

Teacher and Course

One teacher can teach zero or many courses, while each course is assigned to one teacher.

Multiplicity: Teacher (1) — Course (0..*)

Course and Enrollment

One course can have zero or many enrollments, while each enrollment is associated with one course.

Multiplicity: Course (1) — Enrollment (0..*)

System Requirements

The School Management System requires basic hardware and software resources to

operate effectively. The system should be easy to use, reliable, and capable of storing and managing school information.

Hardware Requirements

A computer or laptop.

Sufficient memory and storage space.

Software Requirements

An operating system such as Windows.

A suitable programming environment and database management system.

A UML modelling tool to design and view the class diagram.

Conclusion

In conclusion, the UML class diagram provides a clear representation of the School Management System structure. It identifies the main classes, their attributes and methods, and the relationships between them. This model can help guide the future development of a system that organizes student, teacher, course, and

enrollment information efficiently.

UML Class Diagram

